

Vibration-Based Condition Monitoring and Fault Detection in Rotating Machinery Using Signal Processing and Machine Learning

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1. ABSTRACT

This work presents a vibration based condition monitoring framework for detecting and characterizing mechanical faults in rotating machinery using real experimental data. The analysis uses the Case Western Reserve University (CWRU) bearing dataset, which provides high resolution accelerometer measurements under both normal and faulty conditions.

The objective is to evaluate how time domain analysis, FFT based frequency analysis, and machine learning methods distinguish between healthy and degraded states. The methodology combines statistical feature extraction, spectral analysis, signal segmentation, and unsupervised anomaly detection using the Isolation Forest algorithm.

The results show a clear increase in vibration energy and variability as fault severity increases. RMS values rise from 0.0738 in the normal condition to 0.1392 for a ball fault and 0.2915 for an inner race fault. The ML model detects significantly more anomalous segments in faulty conditions compared to the baseline.

The results confirm that vibration based monitoring can detect early mechanical degradation using computationally efficient methods. The framework is scalable and applicable to real systems such as electric motors, transport infrastructure, and industrial rotating equipment.

2. INTRODUCTION

Reliability of rotating machinery is a key requirement in manufacturing, transportation, and energy systems. Bearings are critical components in these systems, and their failure can result in downtime, increased costs, and safety risks in critical applications.

Mechanical degradation develops gradually through wear, fatigue, misalignment, or localized defects. These processes alter system dynamics and are reflected in vibration behavior. For this reason, vibration analysis remains one of the most effective techniques for condition monitoring and fault detection.

Conventional maintenance strategies rely on scheduled inspections or reactive repair. These approaches do not provide early fault detection and often lead to inefficient maintenance planning. Condition based maintenance (CBM) addresses this by monitoring system signals and identifying degradation before failure occurs.

Accelerometer data can be analyzed in both time and frequency domains to extract features sensitive to faults. RMS, variance, and kurtosis capture energy, variability, and impulsive behavior, while FFT reveals frequency components associated with mechanical defects.

Unsupervised ML methods extend this approach by detecting abnormal behavior without requiring labeled fault data. This is particularly useful in real systems where failure data is limited.

This work develops a practical condition monitoring framework using the CWRU dataset. It combines time domain analysis, FFT, segmentation, and anomaly detection to distinguish between normal and faulty operation, with emphasis on interpretability and real-world applicability.

3. DATASET AND EXPERIMENTAL SETUP

The analysis is based on vibration data from the Case Western Reserve University (CWRU) Bearing Data Center, a standard benchmark for fault diagnosis in rotating machinery.

The experimental setup consists of a motor driven system with rolling element bearings. Accelerometers are mounted at the drive end, fan end, and base. Only drive end data is used in this study, as it provides the most direct measurement of bearing behavior.

Signals are sampled at approximately 12,000 Hz, allowing capture of both low and high frequency components.

Three operating conditions are analyzed:

- 1) Normal condition (97.mat). Represents healthy operation with no defects.

Samples: 243,938

- 2) Ball fault (118 2.mat). Represents a defect on the rolling element.

Samples: 122,571

- 3) Inner race fault (105.mat). Represents a defect on the inner race, typically producing stronger impact behavior.

Samples: 121,265

All signals are converted into one dimensional time series for consistent analysis. The dataset structure enables direct comparison between increasing fault severity.

4. TIME DOMAIN ANALYSIS

Time domain analysis provides a direct view of vibration amplitude over time and highlights irregularities and transient behavior.

The first 2000 samples of each signal were examined.

Normal condition

The signal is stable with low amplitude variation and no significant spikes. This reflects steady operation without localized defects.

Ball fault

The signal shows increased variability and irregular fluctuations. The rolling element defect introduces periodic disturbances, resulting in moderate increases in vibration energy.

Inner race fault

The signal contains high amplitude spikes and irregular patterns. These correspond to repeated impact events caused by the defect on the inner race.

Comparison

A clear progression is observed. The normal condition is stable, the ball fault introduces moderate disturbance, and the inner race fault produces strong impulsive behavior. This reflects increasing fault severity.

5. FEATURE EXTRACTION AND STATISTICAL ANALYSIS

Quantitative analysis was performed using statistical features to distinguish operating conditions.

The following features were extracted:

- RMS
- Standard deviation
- Variance
- Kurtosis

RMS

*measures vibration energy.

- Normal: 0.0738
- Ball fault: 0.1392
- Inner race: 0.2915

Energy increases significantly with fault severity. The inner race condition shows nearly four times the baseline value.

Variance

*reflects signal variability.

- Normal: 0.00528
- Ball fault: 0.0192
- Inner race: 0.0848

The increase indicates growing instability in faulty conditions.

Kurtosis

*captures impulsive behavior.

- Normal: -0.236
- Ball fault: -0.015
- Inner race: 2.396

The high kurtosis value for the inner race fault confirms the presence of impact events.

To sum up, all features show consistent separation between conditions. RMS captures energy growth, variance reflects instability, and kurtosis identifies impulsive faults.

6. FREQUENCY DOMAIN ANALYSIS (FFT)

Time domain analysis shows how vibration amplitude changes over time, but some fault behavior is easier to identify in the frequency domain. Mechanical defects often generate periodic impacts or repeated excitation patterns. These patterns may appear as changes in spectral energy, additional peaks, or harmonic components.

FFT was applied to each vibration signal to convert the data from the time domain into the frequency domain. The analysis used the drive end accelerometer signal and a sampling frequency of 12,000 Hz. Only the positive frequency spectrum was used, since the negative half does not add diagnostic information for real valued signals.

The normal spectrum shows lower energy and a more stable distribution across the selected frequency range. This indicates that the vibration behavior is mainly associated with regular system operation rather than fault induced excitation.

The ball fault spectrum shows increased spectral activity compared to the normal baseline. This suggests that the rolling element defect introduces additional vibration components. The increase is consistent with repeated contact between the damaged rolling element and the bearing surfaces.

The inner race fault spectrum shows the strongest spectral response. This is consistent with the time domain results, where the inner race signal showed high amplitude spikes. The stronger frequency content indicates repeated impacts and more severe mechanical disturbance.

Interpretation: FFT analysis supports the time domain findings. The normal condition remains relatively stable, while the fault conditions show increased vibration energy. The inner race fault produces the strongest response, indicating a more severe fault pattern. This confirms that frequency domain analysis provides useful diagnostic information and strengthens the reliability of the overall condition monitoring framework.

7. SIGNAL SEGMENTATION AND TREND ANALYSIS

A real condition monitoring system does not usually analyze one full signal only once. It processes incoming data continuously in smaller time windows. Segmentation makes the analysis more realistic because it shows how diagnostic indicators change over time.

Each signal was divided into non overlapping windows of 2048 samples. Each window was treated as a short operating interval, and RMS was calculated for each segment.

The segmentation produced:

- Normal condition: 119 segments
- Ball fault condition: 59 segments
- Inner race fault condition: 59 segments

The different number of segments is due to different signal lengths.

Segment level RMS analysis provides a more practical view than a single global RMS value. Instead of showing only one value for the entire signal, it shows how vibration energy changes from segment to segment.

Normal condition: the normal RMS trend remains low and stable across segments. This indicates consistent mechanical behavior and supports the baseline assumption.

Ball fault: the ball fault RMS trend is higher than the normal baseline. The increased values suggest persistent abnormal vibration caused by the rolling element defect.

Inner race fault: the inner race RMS trend is the highest and shows greater variation. This indicates stronger and less stable vibration behavior, consistent with repeated impact events.

Interpretation: segmentation connects the analysis to real monitoring practice. A deployed system would collect sensor data continuously, divide it into windows, calculate features, and detect abnormal trends. This makes the project more than a static dataset comparison. It becomes a prototype logic for continuous monitoring.

8. ANOMALY DETECTION

Anomaly detection identifies when system behavior deviates from a normal baseline. In this project, two methods were used: a threshold based method and an ML based method using Isolation Forest. Using both methods is valuable because it shows the difference between simple rule based monitoring and adaptive data driven monitoring.

8.1 Threshold Based Detection

The threshold method uses the normal RMS value as a reference. The threshold was defined as:

$$\text{Threshold} = 2 \times \text{RMS_normal}$$

Since the normal RMS value was approximately 0.0738, the threshold was approximately 0.1475. Any sample with absolute amplitude above this threshold was counted as anomalous.

The results were:

- Normal condition: 10,344 anomalies
- Ball fault condition: 35,618 anomalies
- Inner race fault condition: 60,649 anomalies

These results show a clear increase in anomaly count across the three conditions. The normal signal still contains some values above the threshold because real vibration data naturally contains fluctuations. However, the fault conditions produce far more anomalies, especially the inner race fault.

This method is simple and easy to explain. It is useful as a baseline monitoring approach, but it depends heavily on manual threshold selection. In real systems, fixed thresholds may need adjustment when operating conditions change.

8.2 ML Based Detection Using Isolation Forest

Isolation Forest was used as an unsupervised anomaly detection model. The model was trained only on RMS segments from the normal condition. This is realistic because normal operating data is usually easier to collect than fault data.

After training on normal behavior, the model was tested on normal, ball fault, and inner race fault segments.

The results were:

- Normal condition: 6 anomalous segments
- Ball fault condition: 59 anomalous segments
- Inner race fault condition: 59 anomalous segments

The model detected very few anomalies in the normal condition and classified nearly all fault segments as anomalous. This shows strong separation between healthy and faulty behavior.

The model does not identify the specific fault type. It only determines whether the segment is normal or abnormal. Fault classification would require additional features or a supervised model. Still, for early warning and maintenance support, this is highly useful.

8.3 Interpretation

The threshold method and Isolation Forest both confirm that faulty conditions deviate clearly from normal behavior. The threshold method is transparent and easy to implement. The ML method is more flexible because it learns baseline behavior automatically.

Together, these results strengthen the project. They show that abnormal vibration can be detected through both classical signal processing logic and modern data driven methods.

9. SYSTEM ARCHITECTURE AND IMPLEMENTATION FRAMEWORK

The analytical pipeline developed in this project can be translated into a real condition monitoring system. The system would use sensors to collect vibration data, process the data in short windows, extract diagnostic features, and generate maintenance alerts when abnormal behavior is detected.

A practical implementation can be structured as follows:

Sensor Layer

Data Acquisition Layer

Signal Processing Layer

Feature Extraction Layer

Anomaly Detection Layer

Decision Support Layer

Sensor layer

Accelerometers collect vibration data from mechanical components. In rotating machinery, these sensors can be mounted on bearing housings, motor casings, gearboxes, or structural supports. In rail applications, they can be installed on turnout mechanisms, switch motors, actuator assemblies, or trackside electromechanical components.

Data acquisition layer

The sensor signal is converted into digital form using data acquisition hardware. Sampling frequency must be high enough to capture relevant mechanical dynamics. In this project, 12,000 Hz data was used, which is suitable for vibration analysis of rotating components.

Signal processing layer

Raw vibration data is prepared for analysis. This may include filtering, normalization, and segmentation into fixed windows. Segmentation is especially important because it allows the system to monitor behavior continuously instead of analyzing only long static signals.

Feature extraction layer

The system computes features such as RMS, variance, kurtosis, and FFT based spectral characteristics. These features reduce complex raw data into interpretable indicators of system condition.

Anomaly detection layer

The extracted features are evaluated using thresholds or ML models. A simple system may use baseline thresholds, while a more advanced system can use Isolation Forest or other unsupervised learning methods.

Decision support layer

The system converts analytical results into practical outputs. These may include anomaly alerts, trend indicators, condition scores, and maintenance recommendations. This layer is important because technical analysis only becomes useful when it supports real operational decisions.

This architecture is practical because it does not require extremely complex computation. Many steps can be performed on edge devices near the sensor, while larger trend analysis can be performed on a central server or cloud platform.

10. PRACTICAL APPLICATIONS AND BENEFITS

The practical value of this project is its applicability to real systems that rely on rotating or electromechanical components. The same logic used in this analysis can support condition monitoring in industrial machinery, transportation infrastructure, power systems, and manufacturing equipment.

1. Electric motors and rotating machinery

Bearings are common failure points in motors and mechanical drives. Detecting abnormal vibration early allows maintenance teams to intervene before failure affects the full system. The RMS, variance, kurtosis, FFT, segmentation, and ML methods used in this project are directly relevant to motor monitoring.

2. Rail infrastructure and turnout systems

This is one of the most important application areas for the proposed endeavor. Rail turnout and switch systems include moving mechanical parts and electromechanical actuators. These components must operate reliably because failures can disrupt service and create safety risks.

A monitoring system based on this project could use accelerometers on switch motors, actuator assemblies, or mechanical linkage components. The system could detect increased vibration, irregular impacts, abnormal energy trends, or repeated anomalies. These indicators may suggest wear, looseness, misalignment, or developing mechanical defects.

3. Energy and power systems

Turbines, generators, pumps, and compressors require reliable rotating components. Fault detection in these systems can prevent downtime and reduce repair costs. The proposed framework is suitable because it can be implemented with low cost sensors and efficient computation.

4. Manufacturing systems

Industrial production depends on machines operating continuously. Unexpected bearing or motor failure can stop production lines. A lightweight condition monitoring system can reduce downtime by identifying abnormal vibration before complete failure.

Benefits

The main benefit is early fault detection. The project shows that faults produce measurable changes in vibration energy, variability, impulsiveness, frequency content, and anomaly patterns. Detecting these changes early supports better maintenance planning.

Another benefit is cost reduction. Condition based maintenance avoids unnecessary scheduled replacement and reduces emergency repairs.

The framework also improves safety and reliability. In transportation and energy systems, failures can have serious consequences. Early warning gives operators time to inspect and repair components before faults become critical.

The approach is scalable. It uses common sensors, interpretable features, and computationally efficient algorithms. This makes it suitable for deployment across many assets without requiring heavy infrastructure.

11. LIMITATIONS OF THE STUDY

This study demonstrates the feasibility of vibration based condition monitoring using real experimental data. Several limitations should be considered.

The dataset was collected under controlled laboratory conditions, so it may not fully represent the noise and variability of operational environments. Only two fault types were analyzed, and the feature set was limited to standard statistical and spectral indicators.

The ML model was used for anomaly detection rather than detailed fault classification. It identifies abnormal behavior but does not determine the exact fault type. The study also does not include physical deployment, long term degradation tracking, or real time system integration.

These limitations do not reduce the value of the work as a proof of concept. They define the next steps required to move from an analytical prototype to an operational monitoring system.

12. FUTURE WORK

Future development can expand this framework in several directions.

The feature set can be improved with more advanced spectral features, envelope analysis, and time frequency methods. These additions may improve sensitivity to specific fault types.

The ML component can also be expanded. Isolation Forest is effective for anomaly detection, but future models could support fault classification and severity estimation.

Long term data collection would allow analysis of degradation trends over time. This would support predictive maintenance and possibly remaining useful life estimation.

A practical next step would be real time implementation using sensors, an embedded data acquisition unit, and a monitoring dashboard. This would validate the framework under realistic operating conditions.

13. CONCLUSION

This project presents a practical framework for vibration based condition monitoring and mechanical fault detection. Using real data from the CWRU dataset, the analysis shows clear differences between normal operation, ball fault, and inner race fault conditions.

Time domain analysis identifies increasing amplitude and impulsive behavior. Statistical features quantify this progression through RMS, variance, and kurtosis. FFT analysis adds frequency domain evidence, while segmentation shows how vibration energy can be tracked over time. Threshold based detection and Isolation Forest both confirm that faulty conditions can be separated from normal behavior.

The project demonstrates that reliable fault indicators can be obtained using efficient and interpretable methods. This makes the framework practical for real world systems where scalability, reliability, and explainability matter.

The proposed approach can support early fault detection, reduce maintenance costs, improve safety, and enable condition based maintenance in industrial and transportation systems. It provides a strong foundation for future development toward real time predictive maintenance applications.